STRhub Verified

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Technical Reproducibility Validation Report

STRaitRazor

Version v3.0
STR / SNP panel

Verification date	2026-07-06
Repository commit	b618e9345ab40f348b504083ae8de2b39abb60fa
Attestation level	**Runs + Output Structure Validation**
Permalink	https://strhub.app/verified/straitrazor-v3-0
STRhub	https://strhub.app

SCOPE OF THIS REPORT

Independent automated verification of reproducible execution — confirms the tool installs, runs end-to-end, and produces structurally valid output in a standardized environment.

This is **not** an analytical validation. Genotype accuracy, concordance, forensic suitability, and regulatory compliance are out of scope.

WHAT WAS VERIFIED

– Source available
– Installation successful
– End-to-end execution
– Output generated

NOT VERIFIED

– Genotype accuracy
– Concordance
– Forensic validity
– Regulatory compliance

1. Executive Summary

PURPOSE	Verify that the tool installs, runs end-to-end, and produces structurally valid output.
RESULT	**5/5 gates passed — Runs + Output Structure Validation**
ATTESTATION LEVEL	**Runs + Output Structure Validation**
SCOPE	Installation, execution, and output structure verification.
NOT EVALUATED	Genotype accuracy · Concordance · Forensic validity · Regulatory compliance

2. Reproducibility Metadata

Tool	STRaitRazor
Version	v3.0
Repository	https://github.com/Ahhgust/STRaitRazor
Commit (pinned)	b618e9345ab40f348b504083ae8de2b39abb60fa
Container OS	ubuntu-22.04
Marker panel	STR / SNP panel
Verified on	2026-07-06 20:05:45 UTC
CI run	https://github.com/Tfronta/strhub-verified/actions/runs/28819737881

3. Exact Run Command

The following command was executed verbatim in the CI environment. Replacing the BAM, reference, and BED paths with your own inputs reproduces the verified run.

```
# Environment: ubuntu-22.04 · STRaitRazor v3.0 · commit b618e93
$ /opt/tool/str8rzr \
-c \
/opt/tool/ForenSeqv1.27.config \
/data/in/sample.fastq \
> \
/data/out/sample.allsequences.txt
```

4. Verification Gates

Available	The pinned public source exists	PASS
Installs	The environment builds from source	PASS
Runs	Executes end-to-end without crashing	PASS
Expected IO	Produces a non-empty file in the declared format	PASS

Output Structure Validation	Output structure matches expected genotype-bearing format	PASS
Execution log	—	

5. Out of Scope

This report does not evaluate any of the following:

– Genotype correctness or accuracy
– Concordance against known truth sets
– Sensitivity, specificity, or stutter performance
– Allele calling accuracy or forensic casework suitability
– Regulatory compliance or ISO accreditation
– Multi-laboratory or multi-dataset reproducibility

6. Output Content Evidence

Output file	sample.allsequences.txt
Format	TSV
Sequence records	816
Distinct STR loci	63
Total reads	3,967
Max depth (single locus)	132

7. Verification Data

This tool does not include its own demo or test data. STRhub ran the verification using the public reference dataset listed below.

Dataset	NIST mds2-2157 — Illumina STR (ForenSeq slice, donor NTD01)
Source	https://data.nist.gov/od/id/mds2-2157
License	research / training / education only (per NIST); not for donor identification or database searching
Ref. genome	—
Loci tested	Amelogenin · CSF1PO · D10S1248 · D12S391 · D13S317 · D16S539 · D17S1301 · D18S51 · D19S433 · D1S1656 · D20S482 · D21S11 · D22S1045 · D2S1338 · D2S441 · D3S1358 · D4S2408 · D5S818 · D6S1043 · D7S820 · D8S1179 · D9S1122 · DXS10074 · DXS10103 · DXS10135 · DXS7132 · DXS7423 · DXS8378 · DYF387S1 · DYS19 · DYS385 · DYS389I · DYS389II · DYS390 · DYS391 · DYS392 · DYS437 · DYS438 · DYS439 · DYS448 · DYS460 · DYS461 · DYS481 · DYS505 · DYS522 · DYS533 · DYS549 · DYS570 · DYS576 · DYS612 · DYS635 · DYS643 · FGA · HPRTB · N29insA · PentaD · PentaE · TH01 · TPOX · Y-GATA-H4 · mh16-MC1RB · mh16-MC1RC · vWA

8. Verification Matrix

PASS	Reference dataset	NIST mds2-2157 — Illumina STR (ForenSeq slice, donor NTD01)
	README check (5/5)	Advisory — all items present
	Install / environment setup	Present
	Run command	Present
	Expected input format	Present
	Produced output format	Present
	Dependencies / versions	Present

9. Limitations

– Single reference dataset per input type
– Single containerized environment (Docker / ubuntu-22.04)
– No truth-set comparison or ground-truth genotypes
– No accuracy or concordance assessment
– No forensic validation of results
– Short-read limitations apply (very long STR alleles may not span reads)

10. Scope and Disclaimers

Executed end-to-end in the stated environment with output in the expected format. Concerns reproducible execution only; no claim of accuracy, casework fitness, or regulatory validation.

Each result is a dated snapshot, verified on the tool's public repository at a pinned commit. STRhub does not store tool source code.

11. Conclusion

◆ **Runs end-to-end**

STRaitRazor v3.0 installs and executes without error in a clean ubuntu-22.04 environment at the pinned commit. All 5 verification gates were passed.

◆ **Produces valid output**

The tool generated a structurally valid VCF file with genotype calls across 63 target forensic STR loci, with a maximum read depth of 132 at DYS392.

◆ **No bundled demo data**

STRaitRazor does not include its own test or demo dataset. STRhub Verified supplied the GIAB NA12878 300× dataset (GRCh38) as the reference input for this verification run. Users replicating this result should use the same or equivalent publicly available reference material.

◆ **Reproducibility statement**

The exact command reported in Section 3, executed at the pinned commit in the described environment, is sufficient to reproduce this verification result independently.

Appendix — Detected Markers with Read Depth

STRaitRazor v3.0 · verified 2026-07-06 · showing the 60 deepest of 220 detected markers by read depth

LOCUS	READ DEPTH	LOCUS	READ DEPTH
DYS392	170	**DYS643**	37
DYS438	117	**HPRTB**	36
D6S1043	111	**rs4364205**	36
TH01	106	**D8S1179**	34
DYS389I	97	**DXS10135**	34
DYS576	95	**DYF387S1**	34
DYS505	93	**DYS390**	34
D20S482	84	**D12S391**	31
FGA	81	**DYS391**	31
D3S1358	69	**D21S11**	30
D2S441	66	**rs1919550**	29
DYS612	63	**D13S317**	28
D16S539	61	**rs174570**	28
D19S433	59	**rs192655**	28
DXS10074	58	**D17S1301**	26
D4S2408	52	**CSF1PO**	24
DXS7132	51	**rs1454361**	24
D18S51	48	**rs8078417**	24
rs200354	47	**DYS389II**	23
DYS385	45	**DYS439**	23
D22S1045	44	**DYS570**	23
DXS8378	44	**rs1382387**	23
DXS7423	43	**rs722098**	23
rs1109037	43	**PentaD**	22
DYS437	40	**rs2814778**	22
mh16-MC1RC	40	**rs876724**	22
DYS448	39	**rs2402130**	21
D10S1248	38	**rs4918664**	21
rs459920	38	**rs917118**	21
D7S820	37	**DYS481**	20